

JavaScript Foundation

Mini Hands-On Activity

Create a small login form:

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Demo</title>
</head>
<body>

  <h2>Login</h2>

  <input type="text" id="username"
    placeholder="Username">

  <br><br>

  <input type="password" id="password"
    placeholder="Password">

  <br><br>

  <button id="loginBtn">Login</button>

  <p id="message"></p>

  <script>

    document.getElementById("loginBtn")
      .addEventListener("click", function() {

      let username =
        document.getElementById("username").value;

      let password =
        document.getElementById("password").value;

      if (username === "admin" &&
        password === "1234") {

        document.getElementById("message")
          .innerText = "Login Successful";

      } else {

        document.getElementById("message")
```

```
        .innerText = "Invalid Login";

    }

});

</script>

</body>
</html>
```

Complete Skill Experiment

Step 1 — Create Project Structure

Create:

```
ecommerce-js
├── index.html
├── register.html
├── login.html
├── style.css
└── script.js
```

Purpose:

File	Purpose
index.html	Home page
register.html	User registration
login.html	User login
style.css	Styling
script.js	JavaScript functionality

Step 2 — Registration Form

Create register.html.

Use fields:

```
Full Name
Email
Mobile Number
Password
```

Confirm Password
Register Button

Example:

```
<!DOCTYPE html>
<html>
<head>
  <title>User Registration</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <h2>Create Account</h2>

  <form id="registerForm">

    <input type="text"
      id="name"
      placeholder="Full Name">

    <input type="email"
      id="email"
      placeholder="Email">

    <input type="text"
      id="mobile"
      placeholder="Mobile Number">

    <input type="password"
      id="password"
      placeholder="Password">

    <input type="password"
      id="confirmPassword"
      placeholder="Confirm Password">

    <button type="submit">
      Register
    </button>

    <p id="message"></p>

  </form>

  <script src="script.js"></script>

</body>
```

```
</html>
```

Step 3 — Capture Form Submission

In script.js:

```
document.getElementById("registerForm")
  .addEventListener("submit", function(event) {

    event.preventDefault();

    console.log("Registration form submitted");

  });
```

Test.

Open browser console:

F12 → Console

Click Register.

Expected:

Registration form submitted

Step 4 — Read Form Values

Add:

```
let name =
  document.getElementById("name").value;

let email =
  document.getElementById("email").value;

let mobile =
  document.getElementById("mobile").value;

let password =
  document.getElementById("password").value;

let confirmPassword =
  document.getElementById("confirmPassword").value;
```

Then:

```
console.log(name);
console.log(email);
console.log(mobile);
```

Explain:

DOM is being used to retrieve information entered by the user.

Step 5 — Required Field Validation

Add:

```
if (name === "" ||
    email === "" ||
    mobile === "" ||
    password === "" ||
    confirmPassword === "") {

    document.getElementById("message")
        .innerText = "Please fill all fields";

    return;
}
```

Test:

- Leave everything empty
- Click Register

Expected:

Please fill all fields

Step 6 — Password Validation

Add:

```
if (password.length < 6) {

    document.getElementById("message")
        .innerText =
            "Password must contain at least 6 characters";

    return;
}
```

Test:

123

Expected:

Password must contain at least 6 characters

Step 7 — Confirm Password

Add:

```
if (password !== confirmPassword) {  
  
    document.getElementById("message")  
        .innerText =  
        "Passwords do not match";  
  
    return;  
}
```

We have used:

- variables
 - strings
 - operators
 - conditions
 - DOM
 - events
 - functions
-

Step 8 — Email Validation

Introduce a simple regular expression:

```
let emailPattern =  
    /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

Then:

```
if (!emailPattern.test(email)) {  
  
    document.getElementById("message")  
        .innerText = "Enter a valid email";  
  
    return;  
}
```

The pattern checks whether the entered text looks like an email address.

Step 9 — Mobile Validation

```
let mobilePattern = /^[0-9]{10}$/;  
  
if (!mobilePattern.test(mobile)) {
```

```
document.getElementById("message")
    .innerText =
    "Enter a valid 10-digit mobile number";

return;
}
```

Step 10 — Create User Object

JSON processing.

Create:

```
let user = {

    name: name,
    email: email,
    mobile: mobile,
    password: password
};
```

Explain:

```
User
↓
JavaScript Object
↓
JSON
↓
Local Storage
```

Step 11 — Convert Object to JSON

Use:

```
let userJSON =
    JSON.stringify(user);
```

Display:

```
console.log(userJSON);
```

result:

```
{"name":"Ravi","email":"ravi@gmail.com","mobile":"9876543210","password":"123456"}
```

Explain:

Object → JSON

```
JSON.stringify(user)
```

JSON → Object

```
JSON.parse(userJSON)
```

Step 12 — Store Registration Data

Local Storage.

```
localStorage.setItem(  
  "registeredUser",  
  userJSON  
);
```

Explain:

Local Storage stores data in the browser and keeps it even after the browser is closed.

Step 13 — Registration Success

Add:

```
document.getElementById("message")  
  .innerText =  
  "Registration successful!";
```

Then:

```
document.getElementById("registerForm").reset();
```

Complete flow:

```
User enters data  
  ↓  
Submit  
  ↓  
Read form values  
  ↓  
Validate  
  ↓  
Create object  
  ↓  
Convert to JSON  
  ↓  
Store in Local Storage  
  ↓  
Registration successful
```

Step 14 — Build Login Page

login.html

```
<!DOCTYPE html>
<html>
<head>
  <title>User Login</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <h2>User Login</h2>

  <form id="loginForm">

    <input type="email"
      id="loginEmail"
      placeholder="Email">

    <input type="password"
      id="loginPassword"
      placeholder="Password">

    <button type="submit">
      Login
    </button>

    <p id="loginMessage"></p>

  </form>

  <script src="script.js"></script>

</body>
</html>
```

Step 15 — Retrieve Local Storage Data

Add:

```
let storedUser =
  localStorage.getItem("registeredUser");
```

Check:

```
console.log(storedUser);
```

Step 16 — Convert JSON Back to Object

```
let user =
```

```
JSON.parse(storedUser);
```

Now we can access:

```
user.email  
user.password  
user.name
```

Explain:

```
Local Storage  
↓  
JSON String  
↓  
JSON.parse()  
↓  
JavaScript Object
```

Step 17 — Validate Login

```
document.getElementById("loginForm")  
  .addEventListener("submit", function(event) {  
  
    event.preventDefault();  
  
    let email =  
      document.getElementById("loginEmail").value;  
  
    let password =  
      document.getElementById("loginPassword").value;  
  
    let storedUser =  
      localStorage.getItem("registeredUser");  
  
    if (storedUser === null) {  
  
      document.getElementById("loginMessage")  
        .innerText =  
          "No registered user found";  
  
      return;  
    }  
  
    let user = JSON.parse(storedUser);  
  
    if (email === user.email &&  
        password === user.password) {  
  
      document.getElementById("loginMessage")
```

```

        .innerText =
        "Login successful";

    } else {

        document.getElementById("loginMessage")
        .innerText =
        "Invalid email or password";
    }

});

```

Step 18 — Introduce Session Storage

difference:

Storage	Purpose
Local Storage	Persistent user data
Session Storage	Data for current browser session

After successful login:

```

sessionStorage.setItem(
    "loggedInUser",
    user.name
);

```

Now the browser remembers that the user is logged in during the session.

Step 19 — Display Logged-In User

In index.html:

```

<h2 id="welcome"></h2>

```

JavaScript:

```

let loggedInUser =
    sessionStorage.getItem("loggedInUser");

if (loggedInUser) {

    document.getElementById("welcome")
    .innerText =
    "Welcome " + loggedInUser;

}

```

result:

Welcome Ravi

Step 20 — Logout

Create:

```
<button onclick="logout()">
  Logout
</button>
```

JavaScript:

```
function logout() {

  sessionStorage.removeItem("loggedInUser");

  alert("Logged out successfully");

  window.location.href = "login.html";
}
```

Explain:

```
Login
↓
Session Storage
↓
User logged in
↓
Logout
↓
Remove Session Storage
```

Step 21 — Error Handling

Introduce:

```
try {

  let user = JSON.parse(storedUser);

} catch (error) {

  console.log("Invalid user data");

}
```

Explain simply:

try



Try to execute code

catch



Handle the error